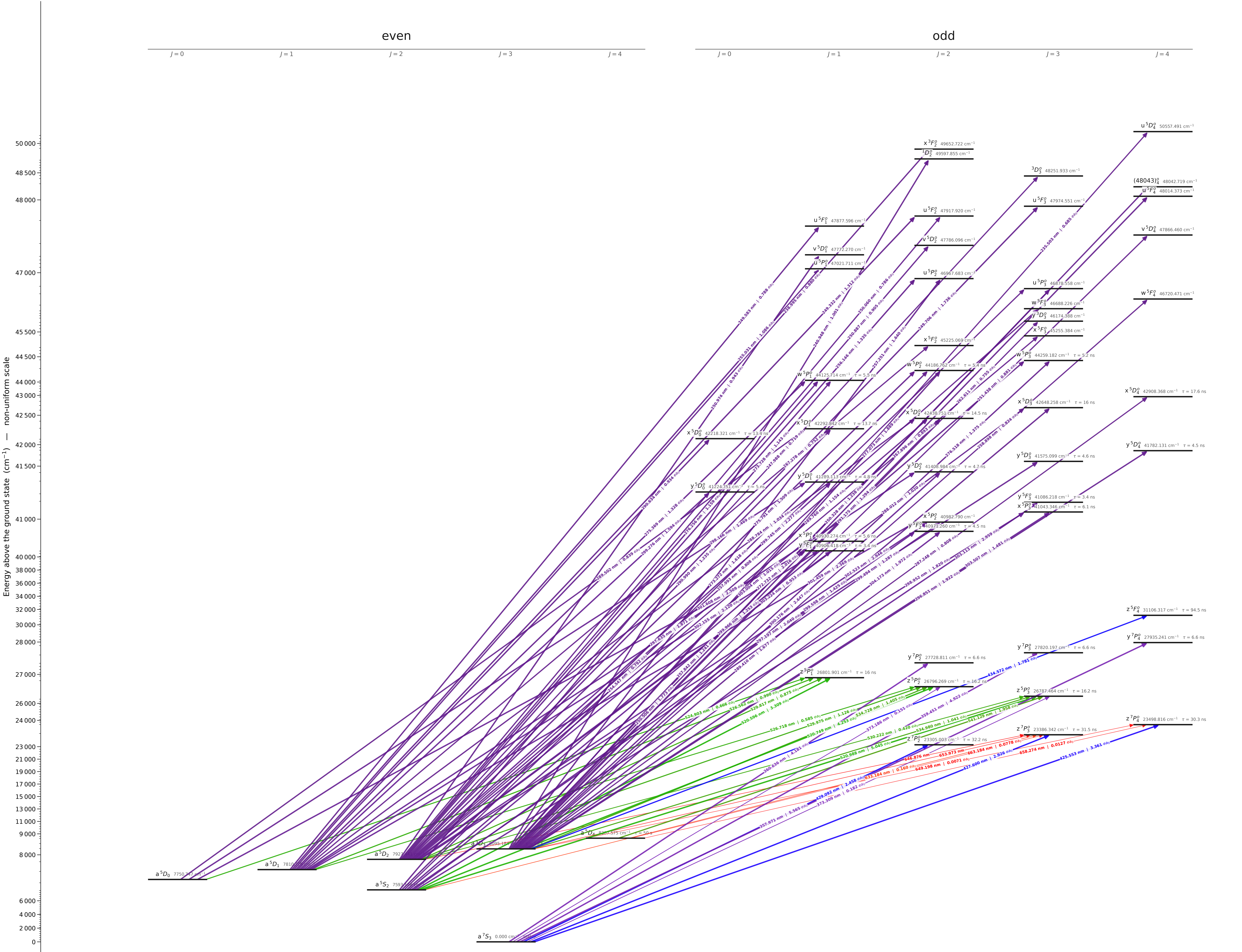




| Key transitions                                                            |            |            |          |                                                       |                      |                |            |                                                                        |
|----------------------------------------------------------------------------|------------|------------|----------|-------------------------------------------------------|----------------------|----------------|------------|------------------------------------------------------------------------|
| Transition                                                                 | λ vac (nm) | λ air (nm) | ν (THz)  | $  \langle U    er    J \rangle  $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                                    |
| a <sup>7</sup> S <sub>3</sub> - y <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 357.9708   | 357.8686   | 837.4775 | 5.565                                                 | 1.520e+08            | 24.11          | 100        | strong UV resonance line a7S3 - y7P4                                   |
| a <sup>7</sup> S <sub>3</sub> - y <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 359.4511   | 359.3485   | 834.0285 | 4.922                                                 | 1.510e+08            | 24.11          | 99.7       | strong UV resonance line a7S3 - y7P3                                   |
| a <sup>7</sup> S <sub>3</sub> - y <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 360.6357   | 360.5329   | 831.2888 | 4.181                                                 | 1.510e+08            | 24.11          | 99.7       | strong UV resonance line a7S3 - y7P2                                   |
| a <sup>7</sup> S <sub>3</sub> - z <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 425.5534   | 425.4336   | 704.4768 | 3.361                                                 | 3.300e+07            | 5.253          | 100        | main cooling line: Zeeman slower, MOT, imaging                         |
| a <sup>7</sup> S <sub>3</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 427.6000   | 427.4797   | 701.1049 | 2.926                                                 | 3.170e+07            | 5.053          | 99.9       | optical pumping to m <sub>J</sub> = -3 and "depumper" towards 5S2 /... |
| a <sup>7</sup> S <sub>3</sub> - z <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 429.0924   | 428.9717   | 698.6664 | 2.458                                                 | 3.100e+07            | 4.943          | 99.8       | 429.1 nm line                                                          |
| a <sup>5</sup> S <sub>2</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 633.1842   | 633.0091   | 473.4680 | 0.160                                                 | 2.900e+04            | 5.053          | 0.09       | 633.2 nm repumper 5S2 -> 7P3                                           |
| a <sup>5</sup> D <sub>2</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 646.8765   | 646.6978   | 463.4462 | -                                                     | -                    | 5.053          | -          | 646.9 nm repumper 5D2 -> 7P3                                           |
| a <sup>5</sup> D <sub>3</sub> - z <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 649.1976   | 649.0182   | 461.7892 | 0.007                                                 | 4.200e+01            | 5.253          | 0          | intercombination leak of the cooling line into metastable...           |
| a <sup>5</sup> D <sub>3</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 653.9727   | 653.7921   | 458.4174 | -                                                     | -                    | 5.053          | -          | 654.0 nm repumper 5D3 -> 7P3                                           |
| a <sup>5</sup> D <sub>4</sub> - z <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 658.2741   | 658.0923   | 455.4219 | 0.013                                                 | 1.270e+02            | 5.253          | 0          | intercombination leak of the cooling line into metastable...           |
| a <sup>5</sup> D <sub>4</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 663.1842   | 663.0011   | 452.0500 | 0.078                                                 | 6.000e+03            | 5.053          | 0.02       | 663.2 nm repumper 5D4 -> 7P3                                           |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

| <sup>53</sup> Cr hyperfine structure (I = 3/2) |              |            |                                                        |                |
|------------------------------------------------|--------------|------------|--------------------------------------------------------|----------------|
| Level                                          | A (MHz)      | B (MHz)    | Shift of each F level from the centre of gravity (MHz) |                |
| a <sup>7</sup> S <sub>3</sub>                  | -83.5990(20) | -          | F=3/2: +501.59                                         | F=5/2: +292.60 |
| a <sup>5</sup> S <sub>2</sub>                  | -173.820     | -          | F=1/2: +782.19                                         | F=3/2: +521.46 |
| a <sup>5</sup> D <sub>3</sub>                  | -35.683(5)   | 15.565(5)  | F=3/2: +223.44                                         | F=5/2: +122.56 |
| a <sup>5</sup> D <sub>4</sub>                  | -48.755(5)   | 63.021(10) | F=5/2: +396.61                                         | F=7/2: +182.64 |
| z <sup>7</sup> P <sub>3</sub> <sup>o</sup>     | -1.5(20)     | -2.000     | F=3/2: +7.80                                           | F=5/2: +5.55   |
| z <sup>7</sup> P <sub>2</sub> <sup>o</sup>     | -11.8(4)     | -0.6(6)    | F=5/2: +88.20                                          | F=7/2: +47.32  |

Measured A, B constants; sources in the data file.

| Isotope shifts                                                             |            |          |             |
|----------------------------------------------------------------------------|------------|----------|-------------|
| Transition                                                                 | λ vac (nm) | Isotopes | Shift (MHz) |
| a <sup>7</sup> S <sub>3</sub> - z <sup>7</sup> P <sub>2</sub> <sup>o</sup> | 425.553    | 53-52    | 10.7(17)    |
| a <sup>7</sup> S <sub>3</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 427.600    | 53-52    | 18(23)      |
| a <sup>5</sup> S <sub>2</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 633.184    | 53-52    | -155(53)    |
| a <sup>5</sup> D <sub>3</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 653.973    | 53-52    | 670(28)     |
| a <sup>5</sup> D <sub>4</sub> - z <sup>7</sup> P <sub>3</sub> <sup>o</sup> | 663.184    | 53-52    | 819(28)     |

Shift = ν(first isotope) - ν(second isotope).